



Foundations of Programming

Assignment No. : 06

PROBLEM STATEMENT: Write a C function to compute the factorial of a number with and without recursion.

OBJECTIVE:

To demonstrate two different approaches for calculating the factorial of a number: **iterative (non-recursive)** and **recursive**, showcasing the versatility of problem-solving methods in programming.

THEORY:

1) Iterative vs Recursive factorial

Iterative	Recursive
Uses loops (for/while)	Function calls itself
Faster, less memory	More memory (stack calls)
Easy to trace	Closer to mathematical definition
No risk of stack overflow	Stack overflow possible for large n

2) Base case in recursion

Base case: **factorial(0) = 1** (or factorial(1)=1).

It stops further function calls. Without base case → infinite recursion → program crash.

3) Handling invalid input (negative numbers)

Factorial is defined only for $n \geq 0$.

Program checks if $n < 0$ and prints error.

This prevents wrong results and infinite recursion.

PLATFORM: Linux

CODE :

```
#include<stdio.h>
```

```
long long fact_iter(int n)
```

```
{
```

```
    long long f=1;
```

```
    for(int i=1;i<=n;i++)
```

```
        f=f*i;
```

```
    return f;
```

```
}
```

```
long long fact_rec(int n)
```

```
{
```

```
    if(n==0 || n==1)
```

```
        return 1;
```

```
    return n*fact_rec(n-1);
```

```
}
```

```
int main()
```

```
{
```

```
    int n;
```

```
    printf("Enter number: ");
```

```
    scanf("%d",&n);
```

```
if(n<0){  
    printf("Factorial not defined for negative numbers");  
    return 0;  
}  
  
printf("Iterative Factorial = %lld\n",fact_iter(n));  
printf("Recursive Factorial = %lld\n",fact_rec(n));  
}
```

INPUT:

```
Enter number: 5
```

OUTPUT:

```
Iterative Factorial = 120  
Recursive Factorial = 120
```

CONCLUSION: Thus, we have successfully executed factorial of a number with and without recursion.

FAQs:

1. What is the purpose of calculating factorial in programming?
2. What is the difference between recursive and non-recursive (iterative) methods?